

Dingxian Cao

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Quantitative Analyst/Researcher

My Website
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Data-driven professional with expertise in banking systems and advanced modeling. Leverages Python, Bash, and cloud computing for model development. Explores GenAI/LLM through personal projects, combining mathematical rigor with programming. Experience in interdisciplinary collaboration and communicating complex technical concepts.

SKILLS

Tools and Languages	Python, JavaScript, Bash, Git, $\text{\LaTeX}$, Markdown, R
Quantitative Research	Statistical Modeling, Economical Modeling, GenAI with LLM, High-dimensional data Modeling
Communication	English, Chinese (native speaker)

WORKING EXPERIENCE

Quantitative Model Validator <i>PNC Bank, NA</i>	Jul 2022 — Present <i>Tysons, VA</i>
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- Quantitative Model Risk Validation.
- GenAI tool risk validation.

GA/RA <i>University of Connecticut</i>	Aug 2017 — May 2022 <i>Storrs, CT</i>
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- Lecturer: Econometrics I(2311), Python for Economists(3322).
- RA role: Prof.Kao, Prof. Choi, Department of Economics.
- TA: Econometrics I (6310), Econometrics III (6310).

RESEARCH PAPERS/PROJECTS

1. Kaggle competition: G-Research Crypto Forecasting <i>Rank: top 1% 18/1946</i>	Nov 2021 — Feb 2022
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- Identify and forecast the non-stationary structure of cryptocurrency markets by *XGBoost*, *LSTM* and *High-dimensional Model*(PhD Thesis).
- Links: [Kaggle account](#), [Github showcase](#), [Empirical case PDF](#)

2. Inference on Panel AR Model with Possible Unit-roots <i>PhD Thesis</i>	Oct 2019 — Jan 2022
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- Uniform Analysis of Dynamic Panel through Local-to-Unity. (submitted to *Econometrica*)
- Asymptotical analysis of persistent Panel model.
- Non-asymptotics approach of building inference on persistent Panel model.
- Non-parametric approach of modeling high-dimensional dependence structure.

3. Global Market Covariance Identification by Multilevel factor model <i>RA project</i>	Oct 2021 — Feb 2022
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- Work as RA of Prof. SungHoon Choi to identify large covariance matrix and pricing factors of index component companies across global markets.

4. Non-asymptotics of High-dimensional Data <i>RA project</i>	2018 — present
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- Work as RA of Prof. Chihwa Kao to study non-asymptotical approaches in Econometrics.
- Field focus: *Concentration Inequalities of Measures*, *Wasserstein distance*, *MinMax Robustness*, *Reproducing kernel Hilbert space*.

EDUCATION

PhD in Economics , <i>University of Connecticut</i>	GPA: 4/4 Aug 2017 - May 2022
M.S. in Statistics , <i>University of Wisconsin-Madison</i>	GPA: 3.7/4 Aug 2015 - May 2017
B.S. in Statistics , <i>East China Normal University</i>	GPA: 3.5/4 Aug 2012 - May 2016

ACTIVITIES/AWARDS

Big Brain Award, PNC Bank	Dec 2023
New England Statistical Symposium 2020	May 2019
Timothy A. and Beverly C. Holt Economics Fellowship	Spring 2019
Graduate School Pre-Doctoral Fellowship	Spring 2021
G200 Youth Forum	May 2015
Cambridge Real Estate and Finance Seminar, CREFS	Jan 2013